

PROVISIONAL PATENT DISCLOSURE

Monocular Spatial Awareness and Object Localization System with Directional Audio Feedback Technical Field

The present disclosure relates to computer vision, mobile augmented reality, and assistive navigation systems, and more specifically to monocular camera-based systems for real-time object detection, spatial localization, distance estimation, and directional guidance without reliance on LiDAR or stereo cameras.

Background and Problem Statement

Modern mobile devices increasingly rely on depth sensors, LiDAR, or stereo camera arrays to estimate object distance and spatial orientation. These requirements limit accessibility due to:

Hardware cost and availability

Increased power consumption

Incompatibility with low-end or mid-range devices

Existing object detection systems typically identify objects only in image space, providing bounding boxes without true spatial context, such as:

Angular direction relative to the user

Estimated real-world distance

Persistent spatial memory across time

Additionally, many systems lack audio-first spatial feedback, limiting usability for visually impaired users or hands-free navigation scenarios.

There exists a need for a single-camera, sensor-assisted system capable of estimating object direction and distance in real-world units, while remaining hardware-agnostic and deployable on standard smartphones.

Summary of the Invention

This disclosure describes a monocular spatial awareness system that integrates:

Real-time object recognition

Motion-based parallax (Structure-from-Motion)

Fiducial and scale anchors

Angular and distance estimation

Persistent object memory

Directional audio feedback

The system enables a mobile device to:

Detect user-specified objects

Estimate their angular bearing and approximate distance

Announce spatial information audibly

Track objects across time and motion

Operate without LiDAR or stereo cameras

System Overview

The system is composed of the following functional layers:

1. Image Acquisition Layer

Continuous monocular RGB video stream from a mobile camera

Optional inertial sensor input (accelerometer, gyroscope)

2. Object Detection Layer

Machine learning-based object recognition using a pre-trained or custom model

Detection outputs include bounding boxes and class labels

User-defined watchlists allow selective tracking of target objects

3. Angular Localization Module

Angular bearing is computed using the horizontal displacement of the object's bounding box center relative to the camera's optical center.

Example:

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```
angle = (x_center / frame_width - 0.5) × horizontal_field_of_view
```

This yields left/right angular orientation in degrees.

4. Distance Estimation Module (Multi-Method)

Distance is estimated using one or more of the following techniques:

a. Apparent Size Estimation

Bounding box area or height inversely correlates with distance

Useful for known object classes or fiducial markers

b. Fiducial Anchors

Objects or markers of known physical size

Provide scale calibration for depth estimation

c. Motion-Based Parallax (Structure-from-Motion)

Device motion creates a virtual stereo baseline over time

Tracked visual features are triangulated across frames

Relative depth is computed using camera pose changes

Scale is injected using IMU data or known anchors

5. Pose and Motion Estimation

Feature tracking (e.g., optical flow)

Outlier rejection via geometric consistency

Visual-Inertial Odometry reduces drift

Camera trajectory is continuously updated

6. Spatial Memory and Persistence

Detected objects are stored with:

Last known angular position

Estimated distance

Timestamp

Enables "last seen" announcements and spatial continuity

7. Audio and User Feedback System

Text-to-speech announces:

Object name

Direction (degrees left/right)

Estimated distance

Feedback is triggered when:

Target objects enter view

User aligns camera with a stored object

Distance thresholds are crossed

Example Use Case

A user specifies a target such as “blue box.”

The system:

Scans the environment using the camera

Identifies candidate objects matching description

Estimates angular bearing and distance

Announces:

“Blue box detected, fifteen degrees right, approximately two meters.”

Implementation Environment

Mobile platforms (Android initially)

Frameworks may include React Native or native mobile stacks

On-device inference preferred for latency and privacy

Backend persistence optional but supported

Novelty and Differentiation

The disclosed system differs from prior art by combining:

Monocular vision without LiDAR

Motion-based depth estimation on consumer hardware

Object-specific spatial memory

Audio-first spatial interaction

User-defined target tracking

into a single integrated pipeline suitable for mobile deployment.

Inventor Information

Inventor: Michael Omondi Oluoch

Residence: Kenya

Inventorship Status: Sole inventor

The inventor declares that he is the original and sole inventor of the system and methods described in this disclosure.

Status of Disclosure

This document is submitted as a Provisional Patent Application and establishes an early priority date for the disclosed invention. It is intended to support future non-provisional and international patent filings.